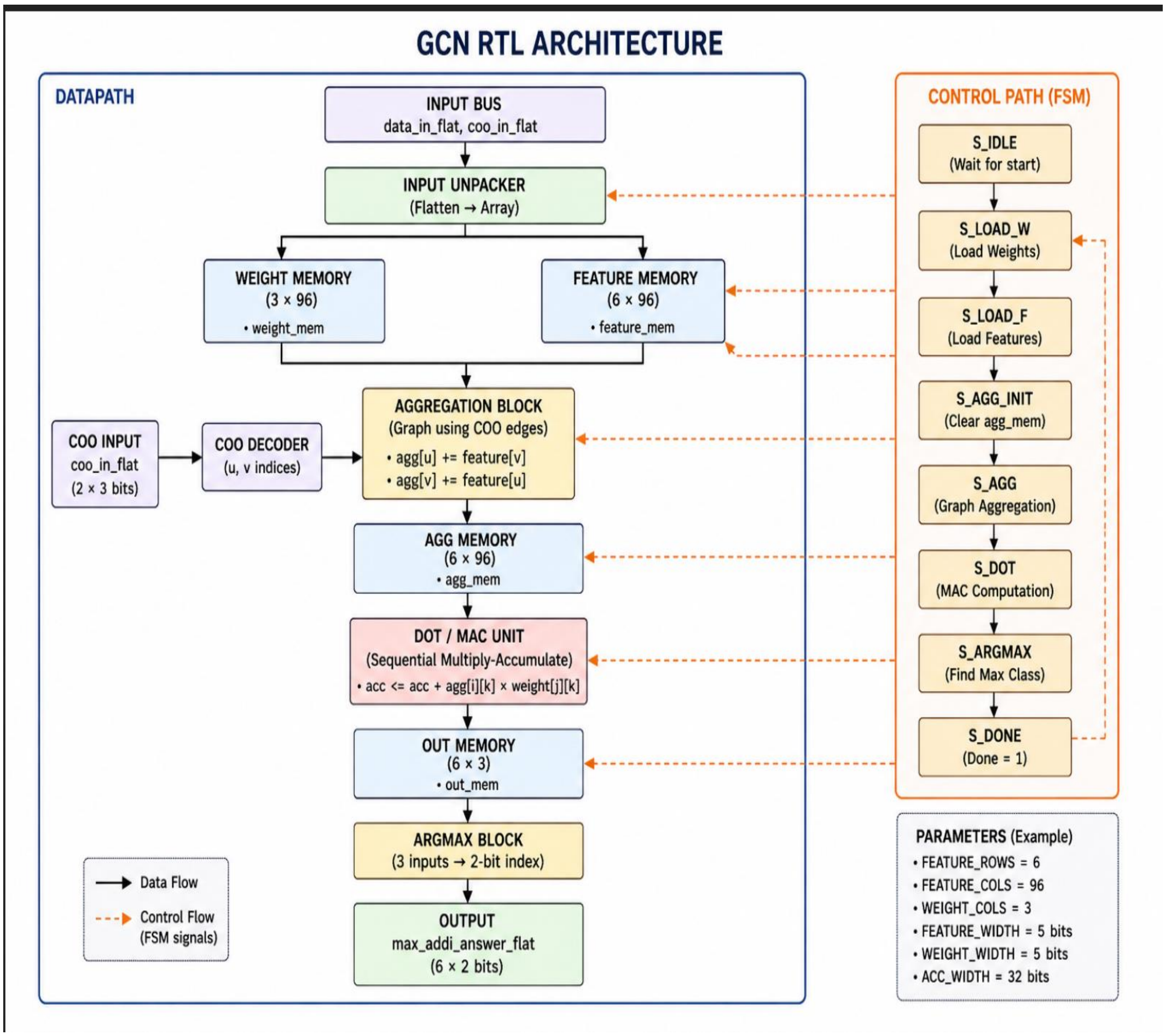
Name: Krish Patel
ASU ID: 1238705171

EEE 525 LAB 4

Milestone 1

Architecture/High-level Block Diagram:

Module (gcn)



High-level block diagram

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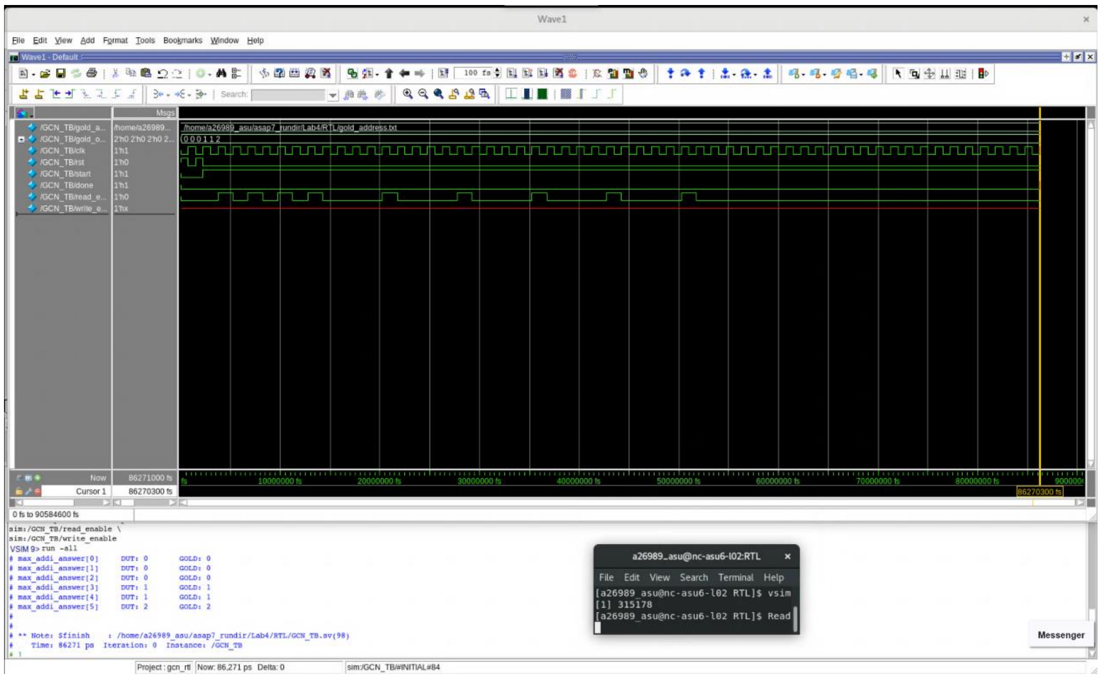
Design decisions: (with 1-2 appropriate figures)

We have designed the GCN module for a node classification application as follows:

- *R*Behavioral Verilog – Simulation:



Correctness of the module with behavioral Verilog

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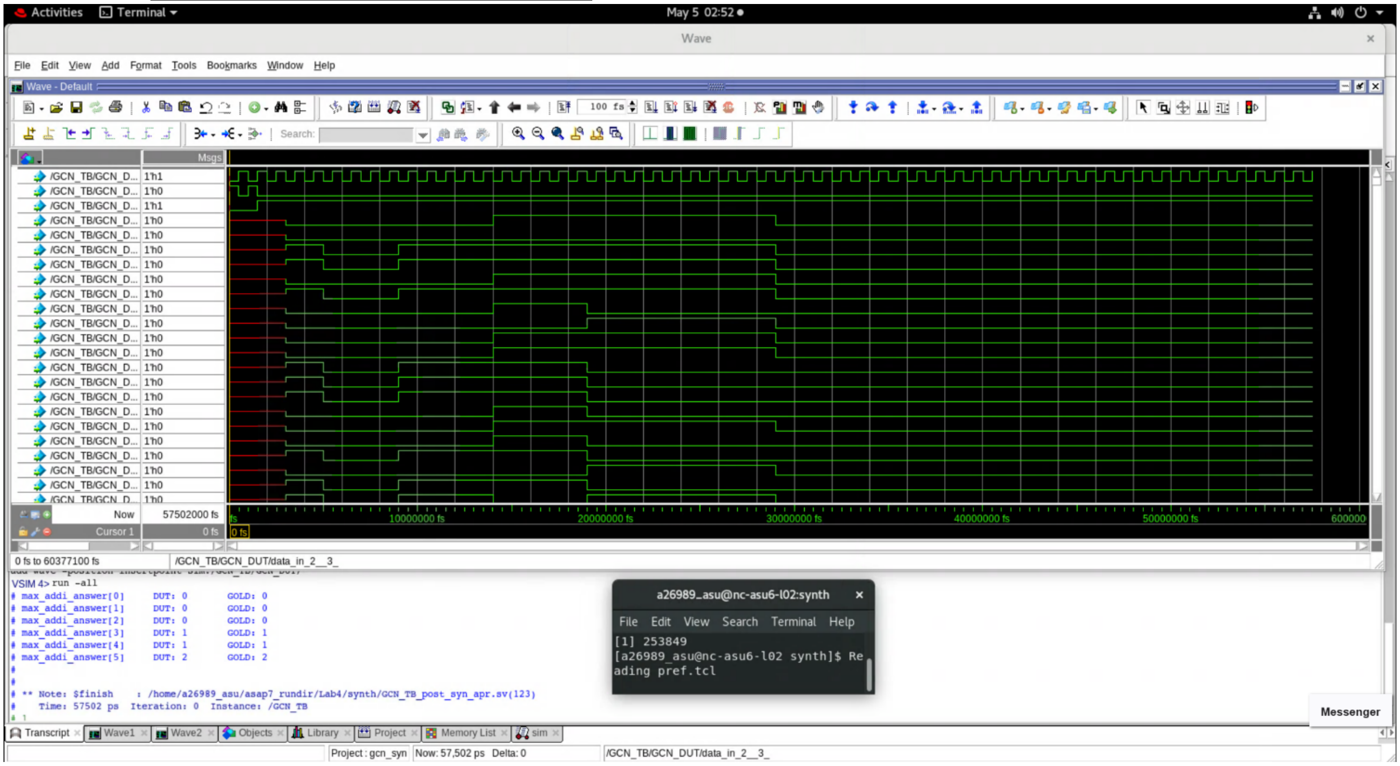
Post_Synthesis – Simulation:



Correctness of the module with post-synthesis Verilog netlist

Milestone 2

Total Latency:

- Total Latency: **86252 ps**, at the clock frequency of **666.67 MHz**.
- Screenshot of Modelsim:

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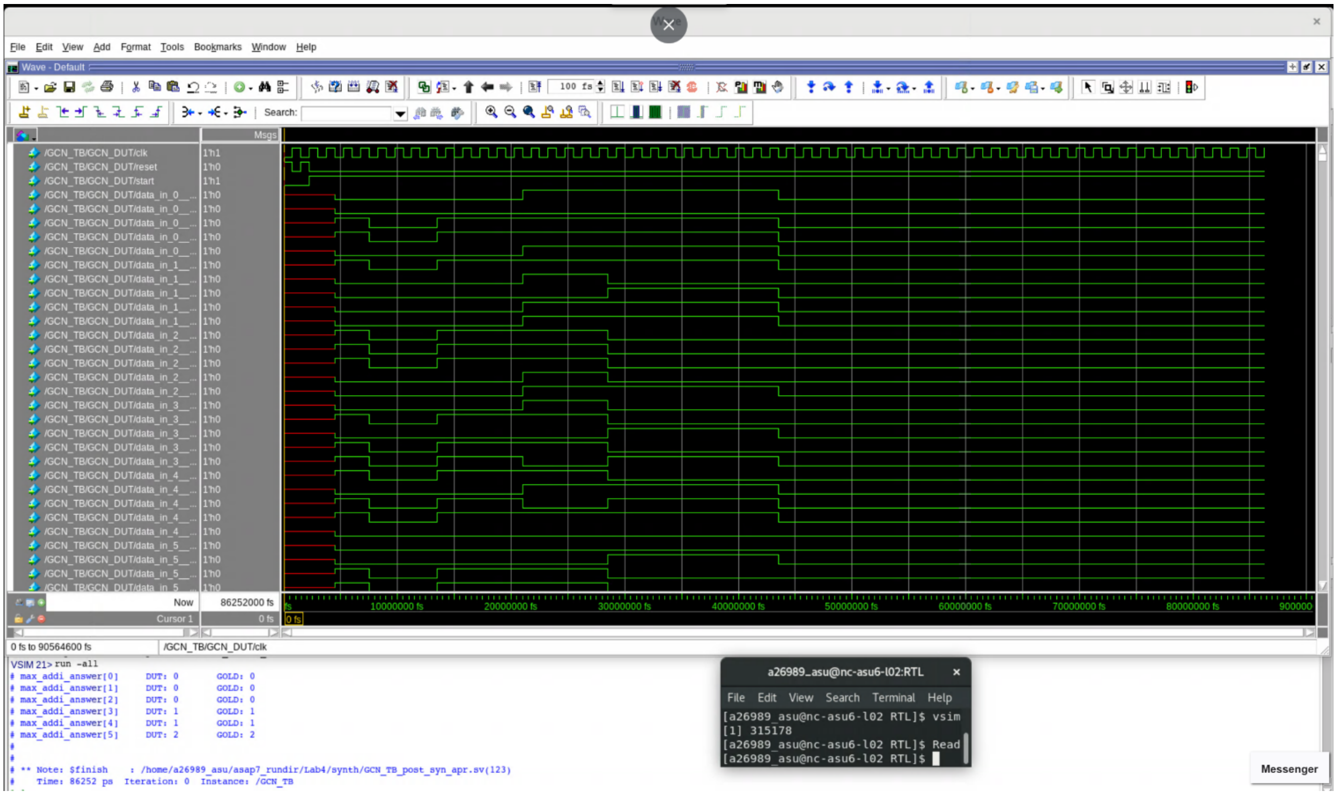
- 

post APR simulations screenshot

Power:

- Total Power (From Innovus): **4.51 mW**

```
-----------------------------------------------------------------------

Total Power
-----------------------------------------------------------------------
Total Internal Power:        2.36988178          52.5082%
Total Switching Power:       2.14110697          47.4394%
Total Leakage Power:         0.00236752           0.0525%
Total Power:                 4.51335629
-----------------------------------------------------------------------
```

```
a26989_asu@nc-asu6-l02:RTL
File  Edit  View  Search  Terminal  Help
[1] 315178
[a26989_asu@nc-asu6-l02 RTL]$ Read
[a26989_asu@nc-asu6-l02 RTL]$
```

Area:

- Standard cells + Filler cells: **0.0483 mm^2**

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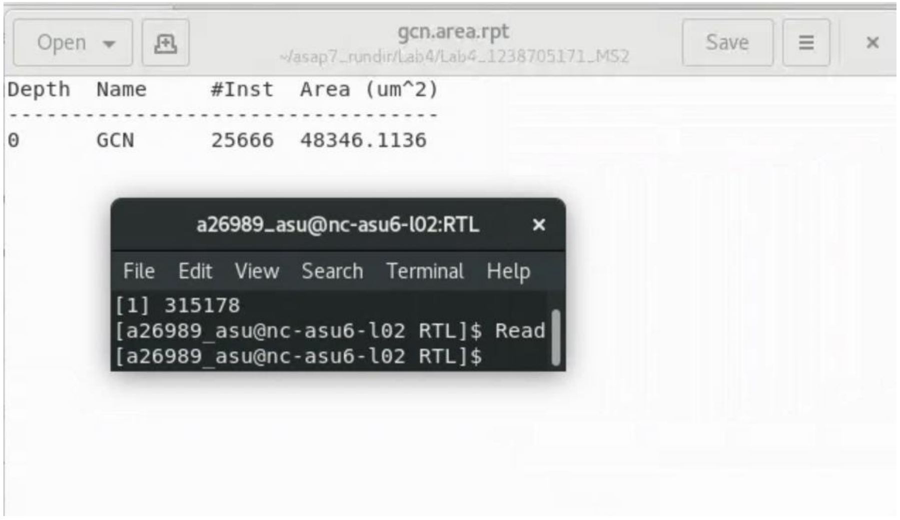
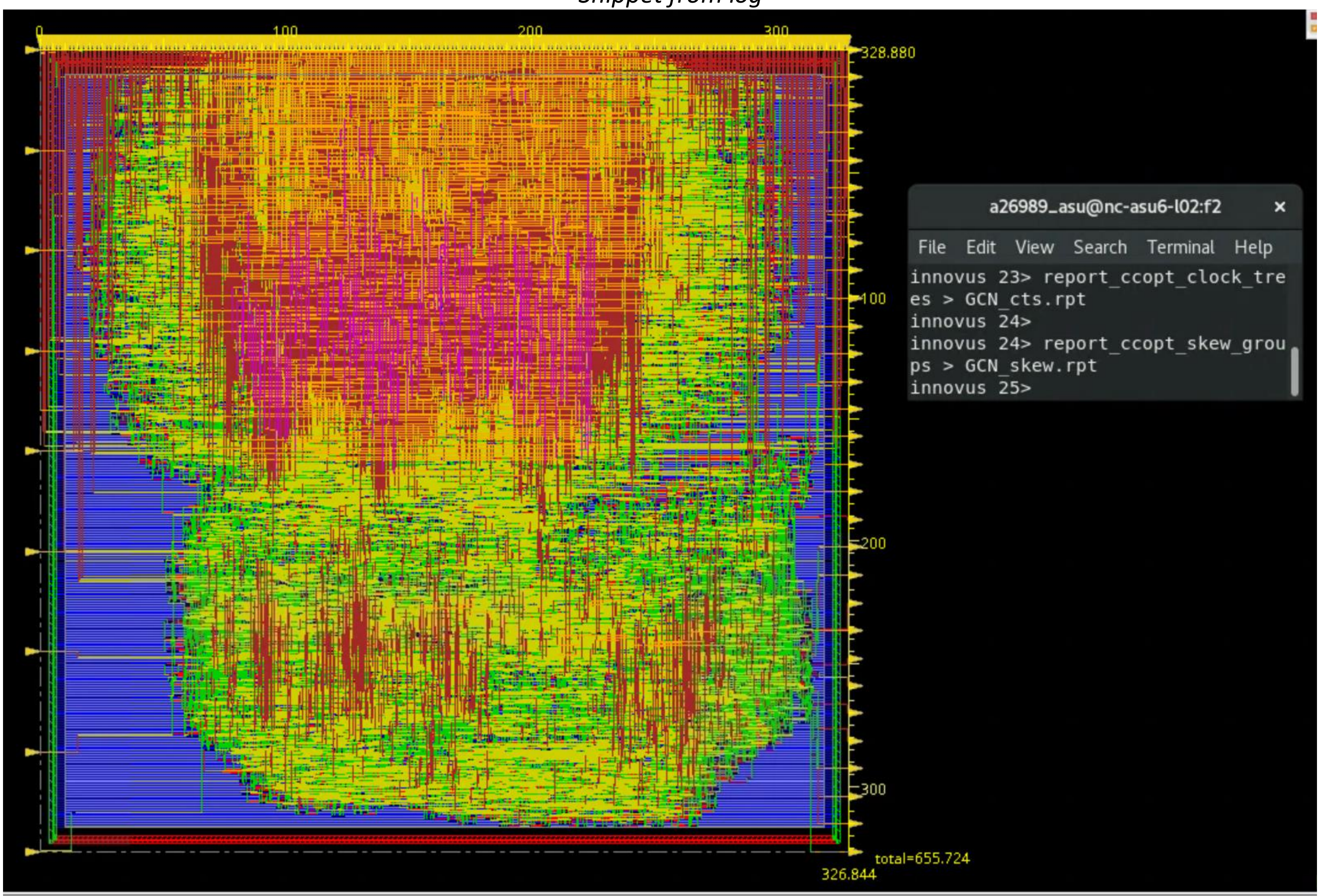


Snippet from log



Design x and y dimensions screenshot

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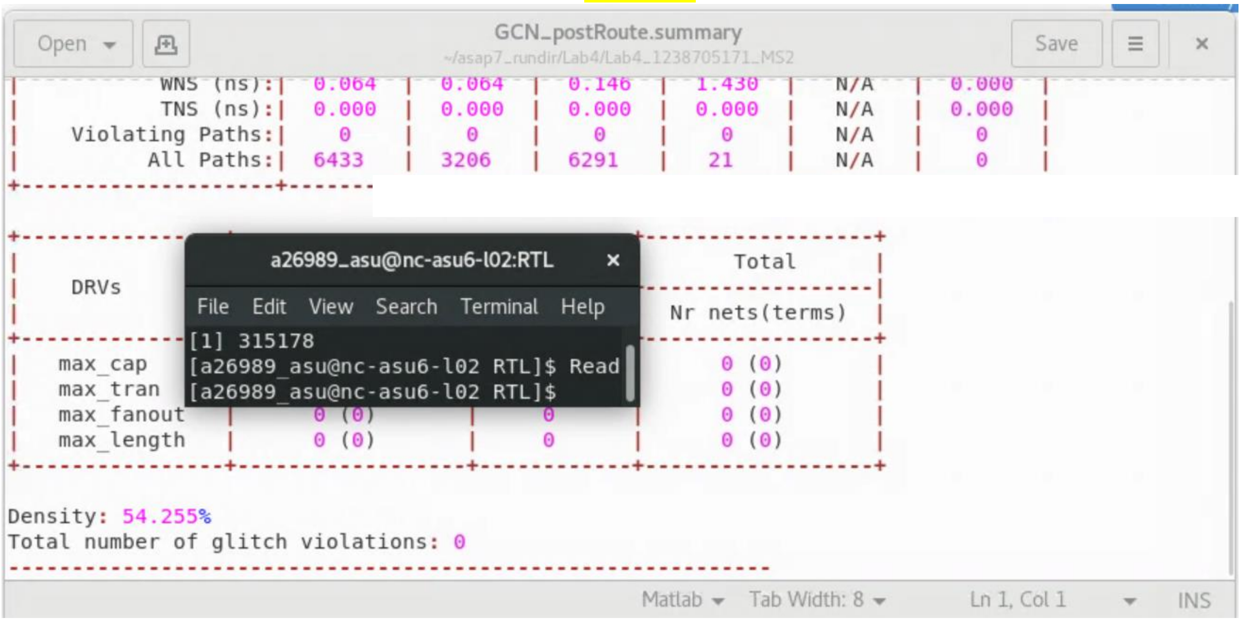
Innovus density:

- Before filler cell insertion: **54.255%**




```
|         WNS (ns):| 0.064  | 0.064  | 0.146  | 1.430  |  N/A   | 0.000  |
|         TNS (ns):| 0.000  | 0.000  | 0.000  | 0.000  |  N/A   | 0.000  |
|  Violating Paths:|   0    |   0    |   0    |   0    |  N/A   |   0    |
|        All Paths:| 6433   | 3206   | 6291   |  21    |  N/A   |   0    |
+--------------------+-------

+----------------                                   +----------------+
|                                                   |     Total      |
|      DRVs                                         +----------------+
|                                                   |  Nr nets(terms)|
+----------------                                   +----------------+
|    max_cap                                        |      0 (0)     |
|    max_tran                                       |      0 (0)     |
|    max_fanout    |     0 (0)        |     0       |      0 (0)     |
|    max_length    |     0 (0)        |     0       |      0 (0)     |
+------------------+------------------+-------------+----------------+

Density: 54.255%
Total number of glitch violations: 0
------------------------------------------------------------
```

```
a26989_asu@nc-asu6-l02:RTL
File  Edit  View  Search  Terminal  Help
[1] 315178
[a26989_asu@nc-asu6-l02 RTL]$ Read
[a26989_asu@nc-asu6-l02 RTL]$
```


Number of gates:

- Gates = **69081**
- Cells = **25666**


```
Gate area 0.6998 um^2
[0] GCN Gates=69081 Cells=25666 Area=48346.1 um^2
```

```
a26989_asu@nc-asu6-l02:RTL
File  Edit  View  Search  Terminal  Help
[1] 315178
[a26989_asu@nc-asu6-l02 RTL]$ Reading pre
[a26989_asu@nc-asu6-l02 RTL]$
```

Post_APR – DRC Check:

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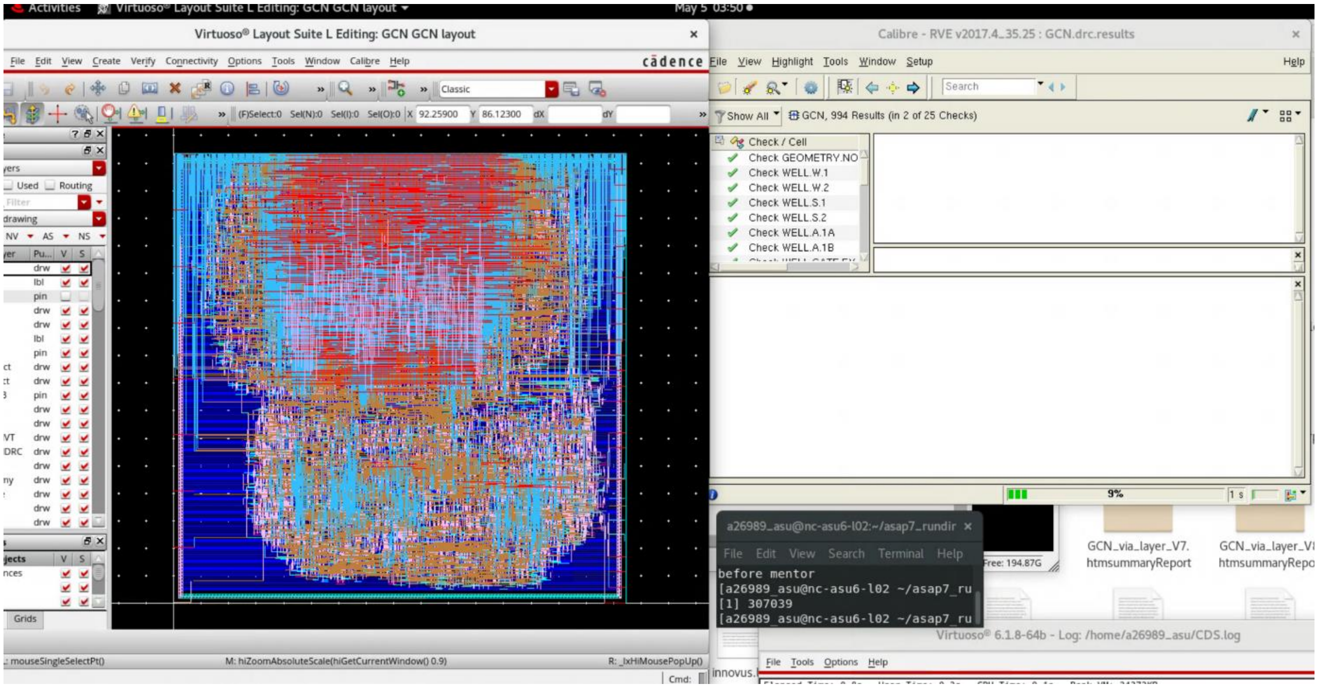
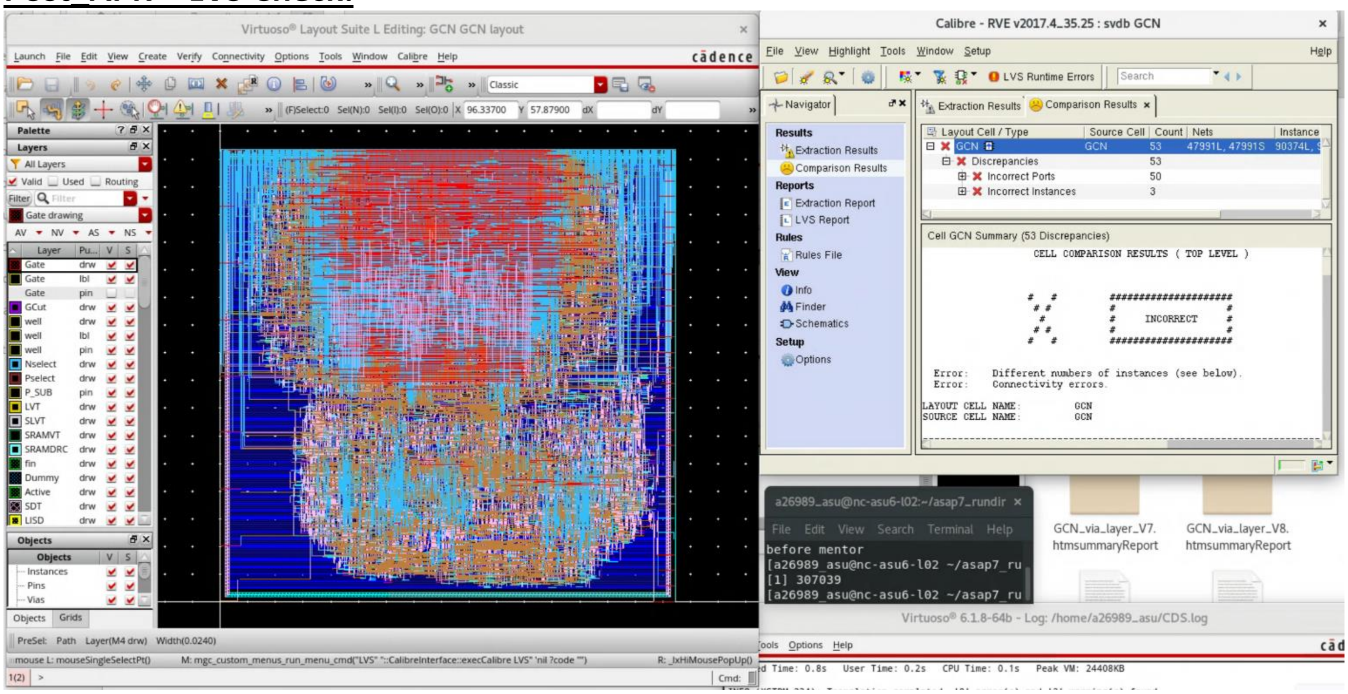


Post_APR – LVS Check:



(Submit your geom.rpt and conn.rpt, as well as attach your screenshot of them here)

Name: Krish Patel
ASU ID: 1238705171

```
gcn.drc.rpt
~/asap7_rundir/Lab4/Lab4_1238705171_MS2
VERIFY DRC ...... Sub-Area: {165.888 221.184 221.184 276.480} 28 of 36  Thread : 1
VERIFY DRC ...... Sub-Area: {55.296 110.592 110.592 165.888} 14 of 36  Thread : 2
VERIFY DRC ...... Sub-Area: {276.480 276.480 328.896 326.880} 36 of 36  Thread : 15
VERIFY DRC ...... Sub-Area: {165.888 110.592 221.184 165.888} 16 of 36  Thread : 9
VERIFY DRC ...... Sub-Area: {0.000 276.480 55.296 326.880} 31 of 36  Thread : 10
VERIFY DRC ...... Sub-Area: {276.480 55.296 328.896 110.592} 12 of 36  Thread : 5
VERIFY DRC ...... Sub-Area: {276.480 165.888 328.896 221.184} 24 of 36  Thread : 11
VERIFY DRC ...... Sub-Area: {110.592 276.480 165.888 326.880} 33 of 36  Thread : 1
VERIFY DRC ...... Sub-Area: {0.000 55.296 55.296 110.592} 7 of 36  Thread : 2
VERIFY DRC ...... Sub-Area: {0.000 165.888 55.296 221.184} 19 of 36  Thread : 6
VERIFY DRC ...... Sub-Area: {221.184 276.480 276.480 326.880} 35 of 36  Thread : 15
VERIFY DRC ...... Sub-Area: {110.592 55.296 165.888 110.592} 9 of 36  Thread : 9
VERIFY DRC ...... Sub-Area: {110.592 165.888 165.888 221.184} 21 of 36  Thread : 7
VERIFY DRC ...... Sub-Area: {221.184 55.296 276.480 110.592} 11 of 36  Thread : 4
VERIFY DRC ...... Sub-Area: {55.296 276.480 110.592 326.880} 32 of 36  Thread : 1
VERIFY DRC ...... Sub-Area: {221.184 165.888 276.480 221.184} 23 of 36  Thread : 11
VERIFY DRC ...... Thread : 3 finished.
VERIFY DRC ...... Sub-Area: {165.888 276.480 221.184 326.880} 34 of 36  Thread : 15
VERIFY DRC ...... Thread : 15 finished.
VERIFY DRC ...... Sub-Area: {55.296 55.296 110.592 110.592} 8 of 36  Thread : 9
VERIFY DRC ...... Thread : 9 finished.
VERIFY DRC ...... Sub-Area: {55.296 165.888 110.592 221.184} 20 of 36  Thread : 7
VERIFY DRC ...... Thread : 7 finished.
VERIFY DRC ...... Sub-Area: {165.888 55.296 221.184 110.592} 10 of 36  Thread : 4
VERIFY DRC ...... Thread : 4 finished.
VERIFY DRC ...... Sub-Area: {165.888 165.888 221.184 221.184} 22 of 36  Thread : 11
VERIFY DRC ...... Thread : 11 finished.

 Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:02:19  ELAPSED TIME: 21.00  MEM: 199.5M) ***

a26989_asu@nc-asu6-l02:RTL
[1] 315178
[a26989_asu@nc-asu6-l02 RTL]$ Read
[a26989_asu@nc-asu6-l02 RTL]$
```


```
gcn.drc.rpt
~/asap7_rundir/Lab4/Lab4_1238705171_MS2
VERIFY DRC ...... Sub-Area: {165.888 221.184 221.184 276.480} 28 of 36  Thread : 1
VERIFY DRC ...... Sub-Area: {55.296 110.592 110.592 165.888} 14 of 36  Thread : 2
VERIFY DRC ...... Sub-Area: {276.480 276.480 328.896 326.880} 36 of 36  Thread : 15
VERIFY DRC ...... Sub-Area: {165.888 110.592 221.184 165.888} 16 of 36  Thread : 9
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VERIFY DRC ...... Sub-Area: {55.296 276.480 110.592 326.880} 32 of 36  Thread : 1
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VERIFY DRC ...... Thread : 15 finished.
VERIFY DRC ...... Sub-Area: {55.296 55.296 110.592 110.592} 8 of 36  Thread : 9
VERIFY DRC ...... Thread : 9 finished.
VERIFY DRC ...... Sub-Area: {55.296 165.888 110.592 221.184} 20 of 36  Thread : 7
VERIFY DRC ...... Thread : 7 finished.
VERIFY DRC ...... Sub-Area: {165.888 55.296 221.184 110.592} 10 of 36  Thread : 4
VERIFY DRC ...... Thread : 4 finished.
VERIFY DRC ...... Sub-Area: {165.888 165.888 221.184 221.184} 22 of 36  Thread : 11
VERIFY DRC ...... Thread : 11 finished.

 Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:02:19  ELAPSED TIME: 21.00  MEM: 199.5M) ***

a26989_asu@nc-asu6-l02:RTL
[1] 315178
[a26989_asu@nc-asu6-l02 RTL]$ Read
[a26989_asu@nc-asu6-l02 RTL]$
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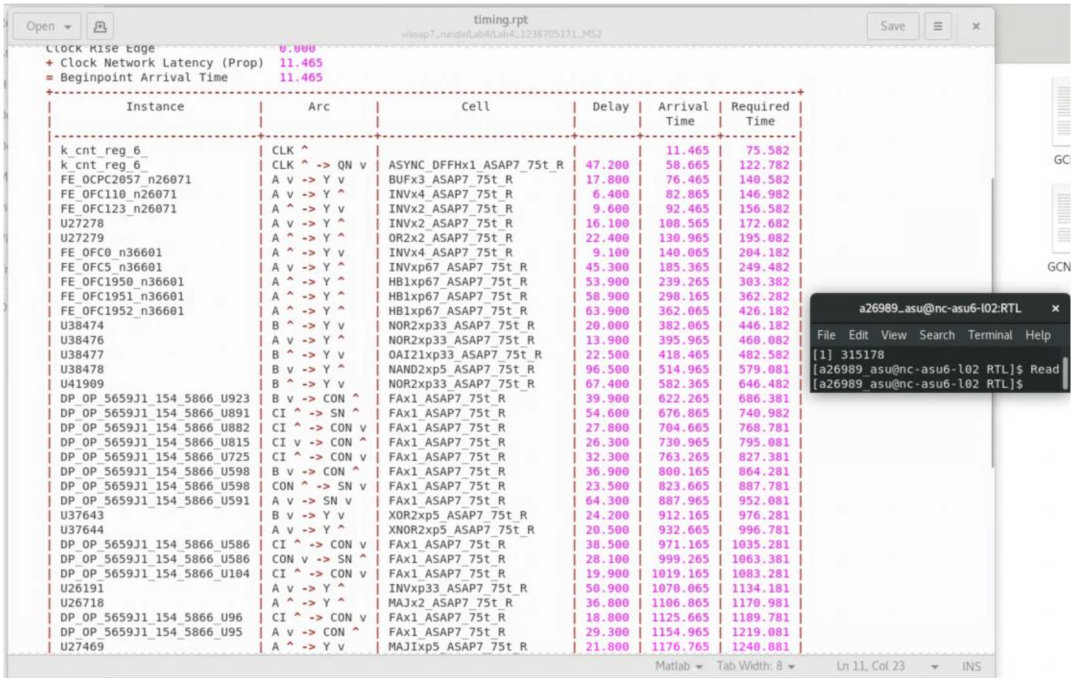

timing.rpt

Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 11.465
= Beginpoint Arrival Time 11.465

Instance	Arc	Cell	Delay	Arrival Time	Required Time
k_cnt_reg_6_	CLK ^			11.465	75.582
k_cnt_reg_6_	CLK ^ -> QN v	ASYNC_DFFHx1_ASAP7_75t_R	47.200	58.665	122.782
FE_OCPC2057_n26071	A v -> Y v	BUFx3_ASAP7_75t_R	17.800	76.465	140.582
FE_OFC110_n26071	A v -> Y ^	INVx4_ASAP7_75t_R	6.400	82.865	146.982
FE_OFC123_n26071	A ^ -> Y v	INVx2_ASAP7_75t_R	9.600	92.465	156.582
U27278	A v -> Y ^	INVx2_ASAP7_75t_R	16.100	108.565	172.682
U27279	A ^ -> Y ^	OR2x2_ASAP7_75t_R	22.400	130.965	195.082
FE_OFC0_n36601	A ^ -> Y v	INVx4_ASAP7_75t_R	9.100	140.065	204.182
FE_OFC5_n36601	A v -> Y ^	INVxp67_ASAP7_75t_R	45.300	185.365	249.482
FE_OFC1950_n36601	A ^ -> Y ^	HB1xp67_ASAP7_75t_R	53.900	239.265	303.382
FE_OFC1951_n36601	A ^ -> Y ^	HB1xp67_ASAP7_75t_R	58.900	298.165	362.282
FE_OFC1952_n36601	A ^ -> Y ^	HB1xp67_ASAP7_75t_R	63.900	362.065	426.182
U38474	B ^ -> Y v	NOR2xp33_ASAP7_75t_R	20.000	382.065	446.182
U38476	A v -> Y ^	NOR2xp33_ASAP7_75t_R	13.900	395.965	460.082
U38477	B ^ -> Y v	OAI21xp33_ASAP7_75t_R	22.500	418.465	482.582
U38478	B v -> Y ^	NAND2xp5_ASAP7_75t_R	96.500	514.965	579.081
U41909	B ^ -> Y v	NOR2xp33_ASAP7_75t_R	67.400	582.365	646.482
DP_OP_5659J1_154_5866_U923	B v -> CON ^	FAx1_ASAP7_75t_R	39.900	622.265	686.381
DP_OP_5659J1_154_5866_U891	CI ^ -> SN ^	FAx1_ASAP7_75t_R	54.600	676.865	740.982
DP_OP_5659J1_154_5866_U882	CI ^ -> CON v	FAx1_ASAP7_75t_R	27.800	704.665	768.781
DP_OP_5659J1_154_5866_U815	CI v -> CON ^	FAx1_ASAP7_75t_R	26.300	730.965	795.081
DP_OP_5659J1_154_5866_U725	CI ^ -> CON v	FAx1_ASAP7_75t_R	32.300	763.265	827.381
DP_OP_5659J1_154_5866_U598	B v -> CON ^	FAx1_ASAP7_75t_R	36.900	800.165	864.281
DP_OP_5659J1_154_5866_U598	CON ^ -> SN v	FAx1_ASAP7_75t_R	23.500	823.665	887.781
DP_OP_5659J1_154_5866_U591	A v -> SN v	FAx1_ASAP7_75t_R	64.300	887.965	952.081
U37643	B v -> Y v	XOR2xp5_ASAP7_75t_R	24.200	912.165	976.281
U37644	A v -> Y ^	XNOR2xp5_ASAP7_75t_R	20.500	932.665	996.781
DP_OP_5659J1_154_5866_U586	CI ^ -> CON v	FAx1_ASAP7_75t_R	38.500	971.165	1035.281
DP_OP_5659J1_154_5866_U586	CON v -> SN ^	FAx1_ASAP7_75t_R	28.100	999.265	1063.381
DP_OP_5659J1_154_5866_U104	CI ^ -> CON v	FAx1_ASAP7_75t_R	19.900	1019.165	1083.281
U26191	A v -> Y ^	INVxp33_ASAP7_75t_R	50.900	1070.065	1134.181
U26718	A ^ -> Y ^	MAJx2_ASAP7_75t_R	36.800	1106.865	1170.981
DP_OP_5659J1_154_5866_U96	CI ^ -> CON v	FAx1_ASAP7_75t_R	18.800	1125.665	1189.781
DP_OP_5659J1_154_5866_U95	A v -> CON ^	FAx1_ASAP7_75t_R	29.300	1154.965	1219.081
U27469	A ^ -> Y v	MAJIxp5_ASAP7_75t_R	21.800	1176.765	1240.881

Screenshot of your timing report for the worst-case hold and setup path

One screenshot of the timing report for the worst setup and worst hold. Do not need the top 3 here.

Design Decision:

- I used a **1500 ps clock period** to ease timing closure and improve slack.
- I increased the **die size** to reduce congestion during placement and routing.
- I implemented **power rings and stripes** to ensure stable and uniform power distribution.
- I placed **inputs on the left and outputs on the right** to simplify routing and improve wire organization.
- I used **higher metal layers (up to M10)** and consistent routing layer constraints during CTS and routing to reduce congestion and routing violations.
- I applied **placement and routing optimizations** to minimize DRC violations and improve overall design quality